

1.3 Show that if $a =_\alpha c$ and $b =_\alpha d$ and both $[b/x]a$ and $[d/x]c$ are defined, then $[b/x]a =_\alpha [d/x]c$

The definition of α -equivalence is given as the strongest congruence satisfying

1. $x =_\alpha x$
2. $o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n) =_\alpha o(\vec{x}'_1.a'_1; \dots; \vec{x}'_n.a'_n)$ if for every $1 \leq i \leq n$, $\hat{\rho}_i(a_i) =_\alpha \hat{\rho}'_i(a'_i)$ for all fresh renamings $\rho_i : \vec{x}_i \leftrightarrow \vec{z}_i$ and $\rho'_i : \vec{x}'_i \leftrightarrow \vec{z}_i$

And substitution for ABTs is defined as

1. $x =_\alpha x$
2. $[b/x]o(\vec{x}_1.a_1; \dots; \vec{x}_n.a_n) = o(\vec{x}_1.a'_1; \dots; \vec{x}_n.a'_n)$ where $\forall i$ we require $\vec{x}_i \notin b$ and we let $[b/x]a_i$ if $x \notin \vec{x}_i$ else $a'_i = a_i$

Let each ABT s expand to $o_s(\vec{x}_{s,1}.a_{s,1}; \dots; \vec{x}_{s,n}).$

Then, $[b/x]a$ expands to $o_a(\vec{x}_{a,1}.a'_{a,1}; \dots; \vec{x}_{a,n}.a'_{a,n})$ with $a'_{a,i} = [b/x]a_{a,i}$ if $x \notin \vec{x}_{a,i}$ else $a'_{a,i} = a_{a,i}$

Similarly $[d/x]c$ expands to $o_c(\vec{x}_{c,1}.a'_{c,1}; \dots; \vec{x}_{c,n}.a'_{c,n})$ with $a'_{c,i} = [d/x]c_{c,i}$ if $x \notin \vec{x}_{c,i}$ else $a'_{c,i} = c_{c,i}$
 Since $a =_\alpha a'$, we know